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Problem A. Coins

Alice and Bob are playing a simple game. They line up a row of n identical coins, all with the heads facing down onto the table and the tails upward.

For exactly m times they select any k of the coins and toss them into the air, replacing each of them either heads-up or heads-down with the same possibility. Their purpose is to gain as many coins heads-up as they can.

Input

The input has several test cases and the first line contains the integer t ($1 \leq t \leq 1000$) which is the total number of cases.

For each case, a line contains three space-separated integers n, m ($1 \leq n, m \leq 100$) and k ($1 \leq k \leq n$).

Output

For each test case, output the expected number of coins heads-up which you could have at the end under the optimal strategy, as a real number with the precision of 3 digits.

Example

standard input	standard output
6 2 1 1 2 3 1 5 4 3 6 2 3 6 100 1 6 100 2	0.500 1.250 3.479 3.000 5.500 5.000

Problem B. The Difference

Alice was always good at math. Her only weak points were multiplication and subtraction. To help her with that, Bob presented her with the following problem.

He gave her four positive integers. Alice can change their order optionally. Her task is to find an order, denoted by A_1, A_2, A_3 and A_4, with the maximum value of $A_1 \times A_2 - A_3 \times A_4$.

Input

The input contains several test cases and the first line provides an integer t ($1 \leq t \leq 100$) indicating the number of cases.

Each of the following t lines contains four space-separated integers.

All integers are positive and not greater than 100.

Output

For each test case, output a line with a single integer, the maximum value of $A_1 \times A_2 - A_3 \times A_4$.

Example

standard input	standard output
5 1 2 3 4 2 2 2 2 7 4 3 8 100 99 98 97 100 100 1 2	10 0 44 394 9998

Problem C. The Number Triangle

As the baby step for learning dynamic programming, the problem about number triangle is very symbolic. Here we will revisit this classic problem.

By starting at any bottom of the triangle below and moving to adjacent numbers on the row above, the maximum total from bottom to top is 23.

3

7 4

2 4 6

8 5 9 3

However, as a lonely outcome of the whole problem, the maximum total is not enough. Similarly, we are concerned with the trail. Consider the new triangle below and a corresponding letter triangle. The maximum total from bottom to top is 6 passing through two "2"(s).

2

1 1

2 1 1

1 1 2 1

a

$a\ b$

$b\ a\ a$

$b\ a\ b\ a$

A best trail can be written by connecting all corresponding letters in the trail consecutively as a string.

Then in the above triangle, the lexicographically smallest best trail starting from the first position of the last row is "bbaa".

The lexicographically smallest best trail starting from the second position of the last row is "abaa".

The lexicographically smallest best trail starting from the third position of the last row is "baaa".

No best trail starts from the fourth position of the last row.

Now, a program is required to determine all positions of the last row leading at least one best trail (a best trail starting from this position). Sort these possible positions by the lexicographical order of the lexicographically smallest best trails which they lead respectively.

Input

The first line of input contains an integer T ($1 \leq T \leq 36$), indicating that there are T test cases.

For each test case, the first line contains an integer N ($N \leq 1500$). The i-th line of the following N lines describes the i-th row of the triangle with n pairs of an integer and a lowercase letter. All integers in input are positive and smaller than 10.

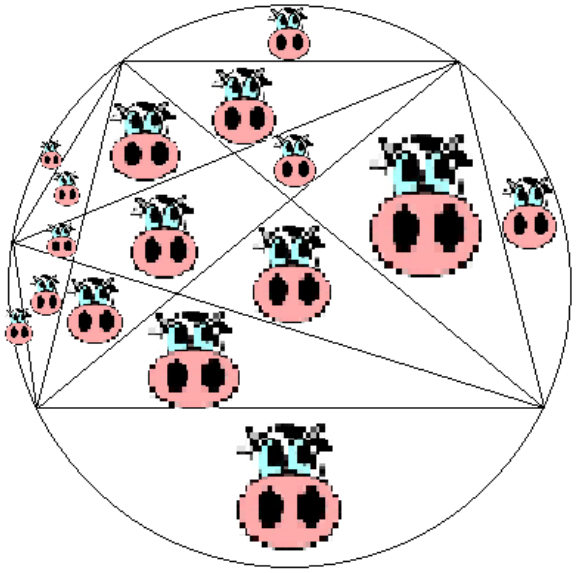
Output

For each query, output a list of numbers in a line indicating the sorted list of possible positions. If two positions own the same lexicographically smallest best trail, output the one with a smaller index in priority order.

Example

standard input	standard output
3 4 3 e 7 v 4 a 2 a 4 o 6 t 8 w 5 f 9 l 3 p 4 2 a 1 a 1 b 2 b 1 a 1 a 1 b 1 a 2 b 2 a 4 2 a 1 a 1 b 2 b 1 a 1 a 1 b 1 a 2 b 1 a	3 4 2 3 1 2 3 1

Problem D. Fence Building



Farmer John owns a farm. He first builds a circle fence. Then, he will choose n points and build some straight fences connecting them. Next, he will feed a cow in each region so that cows cannot play with each other without breaking the fences. In order to feed more cows, he also wants to have as many regions as possible. However, he is busy building fences now, so he needs your help to determine what is the maximum number of cows he can feed if he chooses these n points properly.

Input

The first line contains an integer $1 \leq T \leq 100000$, the number of test cases. For each test case, there is one line that contains an integer n. It is guaranteed that $1 \leq T \leq 10^5$ and $1 \leq n \leq 10^{18}$.

Output

For each test case, you should output a line "Case #i: ans" where i is the test case number, starting from 1 and ans is the remainder of the maximum number of cows farmer John can feed when divided by $10^9 + 7$.

Example

standard input	standard output
3 1 3 5	Case #1: 1 Case #2: 4 Case #3: 16

Problem E. Friends

Some people benefit from large and diverse networks of friends, while others prefer a smaller circle of friends and acquaintances. How about functions.

Consider several functions with coefficients p and μ ($0 \le \mu < p$), denoted by

$$f_0, f_1, \cdots, f_{p-1}.$$

Here p is a prime number and μ is an integer.

The value of function f_i at an integer $x \in \{0, 1, \cdots, p-1\}$ is defined as

$$f_i(x) = ((x^{i+1} \bmod p) + (\mu^{i+1} \bmod p)) \bmod 7.$$

We say a subset S of $\{0, 1, \cdots, p-1\}$ a circle of friends, if all functions in S share the same value at no less than half of positions in $\{0, 1, \cdots, p-1\}$. More specifically, a circle of friends is a subset $S = \{a_1, a_2, \cdots, a_u\}$ of $\{0, 1, \cdots, p-1\}$. It presents u functions $f_{a_1}, f_{a_2}, \cdots, f_{a_u}$, and they have the same value at v distinct locations in $\{0, 1, \cdots, p-1\}$ where $2v \ge p$.

Furthermore, we say a circle of friends is "valuable" if it is maximal. That is say that each bigger set containing a "valuable circle of friends" is not a circle of friends.

Please find and list all "valuable circles of friends".

Input

The inputs contains several test cases. The first line contains an integer T ($1 \le T \le 210$) which is the total number of test cases.

For each test case, a line contains two integers p and μ where p is a prime number and $p \le 100$.

Output

For each test case, if the total number of "valuable circles of friends" is F, output $F + 1$ lines.

Each of the first F lines outputs a subset of $\{0, 1, \cdots, p-1\}$, each of which should be a list of sorted numbers. All valuable circles of friends should be outputted according to the lexicographic order. The last line contains the string "END" as a terminator.

Example

standard input	standard output
2 5 3 7 4	0 4 1 2 3 END 0 3 6 1 4 2 5 END

Problem F. Gathering Minerals

In the game of Starcraft, Terran has 1 idle base, 4 idle workers, 6 supplies and 50 minerals at the beginning.

When a worker is idle, the worker can choose one of the following 3 things to do.

- Gathering minerals: It will cost nothing, and the worker will be busy for 10 seconds. After that, the minerals will increase by 8 and the worker will be idle again.
- Building a supply depot: It will cost 100 minerals immediately, and the worker will be busy for 30 seconds. After that, the supplies will increase by 8 and the worker will be idle again.
- Building a base: It will cost 400 minerals immediately, and the worker will be busy for 120 seconds. After that, a new idle base will be created, and the supplies will increase by 10, and the worker will be idle again.

When a base is idle, it can cost 50 minerals and 1 supply immediately, and it will be busy for 20 seconds. After that, an idle worker will be created and the base will be idle again.

The amount of minerals and supplies cannot be negative all the time. That is, we must have enough minerals and supplies to be cost before doing anything.

Your task is to reach m minerals as soon as possible.

Input

The first line is the number of test cases which is up to 52.

For each test case, there is a line containing only 1 integer m ($m < 2^{64}$).

Output

For each test case, output the minimum time to reach m minerals.

Example

standard input	standard output
3 100 1000 10000	20 200 640

Problem G. The Mountain

All as we know, a mountain is a large landform that stretches above the surrounding land in a limited area. If we as the tourists take a picture of a distant mountain and print it out, the image on the surface of paper will be in the shape of a particular polygon.

From mathematics angle we can describe the range of the mountain in the picture as a list of distinct points, denoted by (x_1, y_1) to (x_n, y_n). The first point is at the original point of the coordinate system and the last point is lying on the x-axis. All points else have positive y coordinates and incremental x coordinates. Specifically, all x coordinates satisfy $0 = x_1 < x_2 < x_3 < \cdots < x_n$. All y coordinates are positive except the first and the last points whose y coordinates are zeroes.

The range of the mountain is the polygon whose boundary passes through points (x_1, y_1) to (x_n, y_n) in turn and goes back to the first point. In this problem, your task is to calculate the area of the range of a mountain in the picture.

Input

The input has several test cases and the first line describes an integer t ($1 \le t \le 20$) which is the total number of cases.

In each case, the first line provides the integer n ($1 \le n \le 100$) which is the number of points used to describe the range of a mountain. Following n lines describe all points and the i-th line contains two integers x_i and y_i ($0 \le x_i, y_i \le 1000$) indicating the coordinate of the i-th point.

Output

For each test case, output the area in a line with the precision of 6 digits.

Example

standard input	standard output
3 3 0 0 1 1 2 0 4 0 0 5 10 10 15 15 0 5 0 0 3 7 7 2 9 10 13 0	1.000000 125.000000 60.500000

Problem H. Count Numbers

Now Alice wants to sum up all integers whose digit sum is exactly a^b.

However we all know the number of this kind of integers are unlimited. So she decides to sum up all these numbers whose each digit is non-zero.

Since the answer could be large, she only needs the remainder when the answer divided by a given integer p.

Input

The input has several test cases and the first line contains the integer t ($1 \leq t \leq 400$) which is the number of test cases.

For each test case, a line consisting of three integers a, b ($1 \leq a, b \leq 20$) and p ($2 \leq p \leq 10^9$) describes the restriction of the digit sum and the given integer p.

Output

For each test case, output a line with the required answer.

Here we provide an explanation of the following sample output. All integers satisfying the restriction in the input are $4, 13, 31, 22, 121, 112, 211$ and 1111. The sum of them all is $4 + 13 + 31 + 22 + 121 + 112 + 211 + 1111 = 1625$ and that is exactly the sample output.

Example

standard input	standard output
5 2 1 1000000 3 1 1000000 2 2 1000000 3 3 1000000 10 1 1000000	13 147 1625 877377 935943

Problem I. A Possible Tree

Alice knows that Bob has a secret tree (in terms of graph theory) with n nodes with $n-1$ weighted edges with integer values in $[0, 2^{60}-1]$. She knows its structure but does not know the specific information about edge weights.

Thanks to the awakening of Bob's conscience, Alice gets m conclusions related to his tree. Each conclusion provides three integers u, v and val saying that the exclusive OR (XOR) sum of edge weights in the unique shortest path between u and v is equal to val.

Some conclusions provided might be wrong and Alice wants to find the maximum number W such that the first W given conclusions are compatible. That is say that at least one allocation of edge weights satisfies the first W conclusions all together but no way satisfies all the first $W+1$ conclusions (or there are only W conclusions provided in total).

Help Alice find the exact value of W.

Input

The input has several test cases and the first line contains an integer t ($1 \le t \le 30$) which is the number of test cases.

For each case, the first line contains two integers n ($1 \le n \le 100000$) and c ($1 \le c \le 100000$) which are the number of nodes in the tree and the number of conclusions provided. Each of the following $n-1$ lines contains two integers u and v ($1 \le u, v \le n$) indicating an edge in the tree between the u-th node and the v-th node. Each of the following c lines provides a conclusion with three integers u, v and val where $1 \le u, v \le n$ and $val \in [0, 2^{60}-1]$.

Output

For each test case, output the integer W in a single line.

Example

standard input	standard output
2 7 5 1 2 2 3 3 4 4 5 5 6 6 7 1 3 1 3 5 0 5 7 1 1 7 1 2 3 2 7 5 1 2 1 3 1 4 3 5 3 6 3 7 2 6 6 4 7 7 6 7 3 5 4 5 2 5 6	3 4

Problem J. Lowest Common Ancestors

In graph theory the lowest common ancestor (LCA) of two nodes v and w in a rooted tree T is the deepest node that has both v and w as descendants. Here we define each node to be a descendant of itself. The LCA of two nodes in T is the shared ancestor of them that is located farthest from the root.

But how about an un-rooted tree?

In this problem you are given an un-rooted tree T with n nodes labelled from 1 to n and several pairs of nodes (v_i, w_i).

For each node x of T, consider the tree rooted by x which becomes a rooted tree; and calculate the summation $\sum_i LCA(v_i, w_i)$.

Input

The input has several test cases and the first line contains an integer t ($1 \le t \le 28$) which is the number of test cases.

For each test case, the first line contains two integers n and q ($1 \le n, q \le 100000$). Each of the following $n-1$ lines describes an edges with two integers v and w ($1 \le v, w \le n$). Then following q lines contains q pairs of nodes (v_i, w_i) described as above ($1 \le v_i, w_i \le n$).

Both of the sum of n and the sum of q in input are smaller than 1000000.

Output

For each test case, output a line with n integers.

The i-th one is the value of $\sum_i LCA(v_i, w_i)$ corresponding to the tree rooted by the i-th node.

Example

standard input	standard output
2 4 3 1 2 1 3 1 4 2 3 3 4 2 4 4 1 1 2 2 3 3 4 1 4	3 5 7 9 1 2 3 4

Problem K. Sum of the Line

Consider a triangle of integers, denoted by T. The value at (r, c) is denoted by $T_{r,c}$, where $1 \le r$ and $1 \le c \le r$. If the greatest common divisor of r and c is exactly 1, $T_{r,c} = c$, or 0 otherwise.

Now, we have another triangle of integers, denoted by S. The value at (r, c) is denoted by $S_{r,c}$, where $1 \le r$ and $1 \le c \le r$. $S_{r,c}$ is defined as the summation $\sum_{i=c}^{r} T_{r,i}$.

Here comes your turn. For given positive integer k, you need to calculate the summation of elements in k-th row of the triangle S.

Input

The first line of input contains an integer t ($1 \le t \le 10000$) which is the number of test cases.

Each test case includes a single line with an integer k described as above satisfying $2 \le k \le 10^8$.

Output

For each case, calculate the summation of elements in the k-th row of S, and output the remainder when it divided by 998244353.

Example

standard input	standard output
2 2 3	1 5